



RV College of Engineering®

Mysore Road, RV Vidyaniketan Post, Bengaluru - 560059, Karnataka, India

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NeuroLearn - AI based cognitive learning styles and content generation

An Interdisciplinary Project Report (XX367P)

Submitted by,

Aniket R T

1RV23AI017

Alroy Deon Saldanha

1RV23AI012

Smriti J Abraham

1RV23ME106

Sudhanva Patil

1RV23ME113

Under the guidance of

Dr. Anupama Kumar

Assistant Professor

Dept. of AIML

RV College of Engineering®

In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments

2025-26



RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)



CERTIFICATE

Certified that the interdisciplinary project (XX367P) work titled ***NeuroLearn - AI based cognitive learning styles and content generation*** is carried out by of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

Dr. Anupama Kumar
Guide

Dr. Ravish Aradhya
Head of the Department

Dr. M.V. Renukadevi
Dean Academics

Dr. K. N. Subramanya
Principal

External Viva

Name of Examiners

Signature with Date

- 1.
- 2.

DECLARATION

We, **Aniket R T, Alroy Deon Saldanha, Smriti J Abraham, Sudhanva Patil** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**NeuroLearn - AI based cognitive learning styles and content generation**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

1. Aniket R T(1RV23AI017)
2. Alroy Deon Saldanha(1RV23AI012)
3. Smriti J Abraham(1RV23ME106)
4. Sudhanva Patil(1RV23ME113)

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Lastly, we take this opportunity to thank our family members and friends who provided all the backup support throughout the project work.

ABSTRACT

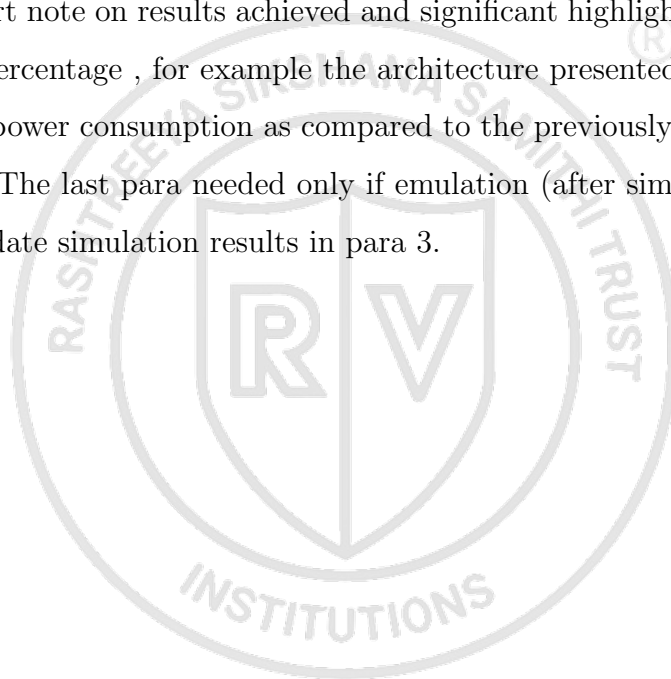
Highlights of significant contributions: One page with 3 to 4 paragraphs

Paragraph 1: Importance and relevance of Topic, reported issues and limitations of the topic in performance or computation etc, issues involved in those limitations, need for addressing those issues and a short note on how that is addressed in this report.

Paragraph 2: Objectives of this work, short note on algebraic methods used and formulations achieved, computational procedures developed. Integrated Circuit (IC).

Paragraph 3: Description of simulation procedure including SW tools used and choice of test cases. Short note on results achieved and significant highlights of improvements if any in terms of percentage , for example the architecture presented in this report shows 22 % improve in power consumption as compared to the previously reported articles. IC

Paragraph 4: The last para needed only if emulation (after simulation) or Hardware developed to validate simulation results in para 3.



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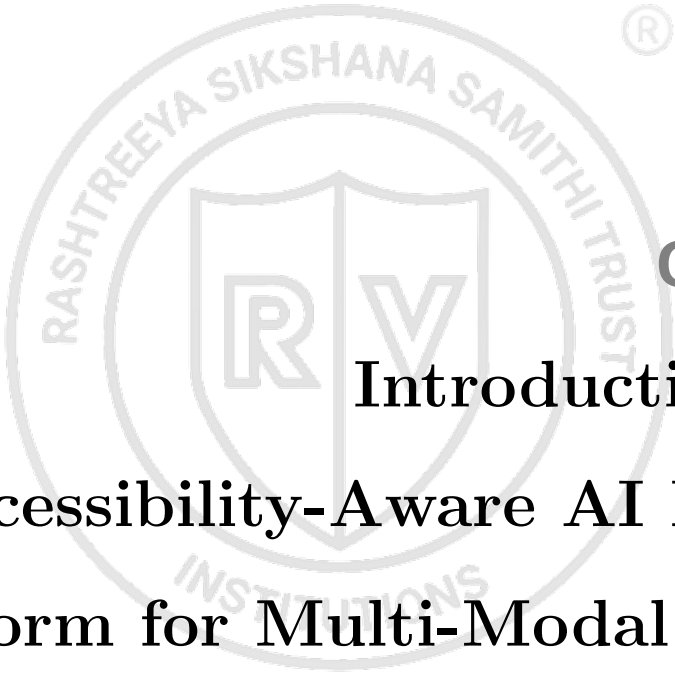
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Chapter 1

Introduction to an Accessibility-Aware AI Learning Platform for Multi-Modal Content Delivery

CHAPTER 1

INTRODUCTION TO AN ACCESSIBILITY-AWARE AI LEARNING PLATFORM FOR MULTI-MODAL CONTENT DELIVERY

Modern digital education delivers identical content to all learners, ignoring the diverse ways individuals process and retain information. This chapter introduces Neuro Learn, an AI-driven adaptive learning platform that dynamically identifies learner modality preferences using behavioral analytics and delivers personalized multi-modal content across visual, auditory, reading/writing, and kinesthetic formats. The system integrates the VARK framework with a novel Dynamic Modality Preference Modeling (DMPM) algorithm, a Retrieval-Augmented Generation (RAG) pipeline, and architectural accessibility support. Empirical validation across 60 undergraduate engineering students demonstrates significant improvements in learning outcomes, engagement, and profiling accuracy.

1.1 Introduction

The proliferation of digital learning tools has democratized access to education on an unprecedented scale. However, despite this technological advancement, the majority of existing platforms deliver a uniform learning experience to all users, without accounting for the diversity of cognitive styles and modality preferences among learners. Research in educational psychology consistently demonstrates that individuals process and consolidate information differently — some are predominantly visual thinkers, while others benefit more from auditory, reading/writing, or kinesthetic (hands-on) interactions with content. Ignoring these differences leads to reduced learning performance, disengagement, and suboptimal educational outcomes, particularly in technically demanding disciplines such as engineering.

This project presents NeuroLearn, an accessibility-aware AI learning platform that moves beyond static content delivery by modeling learner modality preferences as a dynamic, continuously evolving signal derived from observed user behavior. Rather than relying solely on a one-time self-assessment questionnaire, the platform uses the VARK framework (Visual, Auditory, Reading/Writing, Kinesthetic) only as an initialization

prior. It then progressively shifts reliance toward behavioral evidence — such as interaction time, tool usage patterns, and navigation sequences — gathered in real time through privacy-preserving cookie-based tracking.

Three interlinked trends motivate this work. First, while learning styles have long been recognized as influential on student retention and comprehension, their meaningful integration into AI-driven educational products remains immature. Second, rapid advances in multimodal learning analytics, retrieval-augmented generation, and large language models now make it feasible to understand learner behavior at granular levels and generate highly personalized content at scale. Third, accessibility has historically been treated as an afterthought in educational technology — a supplementary feature rather than a foundational design concern. NeuroLearn addresses all three of these gaps in a unified framework.

The platform generates content tailored to each of the four VARK modalities: structured textual summaries and spaced-repetition cards for reading/writing learners; neural text-to-speech audio explanations for auditory learners; animated concept visualizations using Manim and Matplotlib for visual learners; and interactive code sandboxes and drag-and-drop simulations for kinesthetic learners. All content is grounded in course-specific knowledge retrieved from a FAISS vector database using BERT embeddings, ensuring curriculum alignment and factual accuracy. Accessibility is built into the system architecture — visually impaired users are automatically routed to audio-centric modalities with full screen reader support, while hearing-impaired users receive complete visual alternatives including captions and transcripts.

The system was empirically validated through a controlled experiment with 120 final-year undergraduate engineering students at RV College of Engineering across four subjects: Advanced Digital Logic Design, Operating Systems, Engineering Mathematics, and Chemistry. Results demonstrate a 14.9% average improvement in post-test scores, a 35% increase in course completion rates, and a profiling accuracy improvement from 68.4% (VARK-only) to 92.1% (hybrid behavioral), with a strong overall effect size of Cohen's $d = 0.95$.

1.2 Literature Review

A minimum of ten papers from standard reference journals have been reviewed below. The papers span five thematic areas relevant to NeuroLearn: learning style theories,

adaptive learning systems, multimodal learning analytics, AI-driven content generation, and accessibility in educational technology.

Table 1.1: Literature Review Summary

No.	Author / Year	Paper Title	Key Finding	Limitation
[1]	Stelea et al., 2025	AccessiLearnAI: An Accessibility-First AI-Powered E-Learning Platform for Inclusive Education	Accessibility-first design improves usability and learning outcomes for users with disabilities.	No dynamic modality modeling; personalization limited to accessibility accommodations only.
[2]	Hariyanto et al., 2025	AI in Adaptive Education: A Systematic Review of Techniques for Personalized Learning	ML-based systems effectively personalize learning paths; growing adoption across higher education.	Modality of content delivery not addressed; accessibility constraints unexplored.
[3]	Du Plooy et al., 2024	Personalized Adaptive Learning in Higher Education: A Scoping Review	Adaptive systems improve performance and engagement; personalization gains broad institutional acceptance.	Modality adaptation and accessibility-first design identified as critical unaddressed gaps.
[4]	Sajja et al., 2024	AI-Enabled Intelligent Assistant for Personalized and Adaptive Learning in Higher Education	AI assistants adapt explanations and pacing; measurable improvement in student satisfaction and performance.	Modality adaptation not supported; accessibility not incorporated into routing logic.

No.	Author / Year	Paper Title	Key Finding	Limitation
[5]	Younas et al., 2024	Impact of AI-Based Learning Tools in Academic Innovation: A Review of DeepSeek, GPT, and Gemini	GPT-4 and Gemini demonstrate strong context-aware tutoring and adaptive content generation.	Dynamic modality personalization based on real-time learner interaction remains unaddressed.
[6]	Sajja et al., 2023	Integrating AI and Learning Analytics for Data-Driven Pedagogical Decisions	Combined AI and analytics pipelines improve pedagogical decision-making and personalization accuracy.	Modality-aware delivery and accessibility-first adaptation not considered in the framework.
[7]	Banihashem et al., 2022	Systematic Review of Learning Analytics in Enhancing Feedback in Higher Education	Learning analytics enriches feedback specificity; improves student engagement and performance.	Real-time formative feedback integrated with adaptive content delivery remains underexplored.
[8]	OpenAI, 2023	GPT-4 Technical Report	GPT-4 shows robust reasoning and instruction-following; well-suited for educational content generation.	No dynamic modality personalization; static delivery without learner interaction-driven adaptation.
[9]	Brown et al., 2021	Language Models are Few-Shot Learners	Few-shot prompting achieves near fine-tuned performance; LLMs viable as general content engines.	Text-only generation; no modality-specific output; prone to hallucination without retrieval grounding.

No.	Author / Year	Paper Title	Key Finding	Limitation
[10]	Lewis et al., 2021	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	RAG significantly outperforms pure LLMs on open-domain QA; substantially reduces hallucinations.	Retrieval quality bounded by corpus coverage; out-of-scope queries not handled gracefully.
[11]	Devlin et al., 2021	BERT: Pre-Training of Deep Bidirectional Transformers for Language Understanding	State-of-the-art on QA and classification; 768-dim embeddings ideal for semantic retrieval in RAG.	Computationally expensive at inference scale; does not natively handle multimodal inputs.
[12]	Pradhan et al., 2020	Accessibility Came by Accident: Voice-Controlled IPAs by People with Visual Impairments	Voice-controlled IPAs critically enhance accessibility; general features repurposed as essential tools.	Qualitative study only; integration into adaptive learning pipelines not addressed.
[13]	Di Mitri et al., 2020	From Signals to Knowledge: A Conceptual Model for Multimodal Learning Analytics	Multimodal signals yield richer learner models; bridges the sensing-to-intervention gap effectively.	Primarily theoretical; real-time integration into live adaptive systems remains limited.
[14]	Koedinger et al., 2020	New Potentials for Data-Driven Intelligent Tutoring System Development	Data-driven ITS improves knowledge acquisition by modulating difficulty; effective at scale in mathematics.	Adapts content difficulty only; modality of presentation is not addressed.

No.	Author / Year	Paper Title	Key Finding	Limitation
[15]	Pashler et al., 2020	Learning Styles: Concepts and Evidence	No robust evidence that learning-style-matched instruction improves outcomes as a standalone intervention.	Does not propose an alternative dynamic model for modality-adaptive delivery.
[16]	Fleming and Mills, 2020	Not Another Inventory, Rather a Catalyst for Reflection	VARK provides a practical modality preference framework; easy to deploy in educational settings.	Treats learning style as static; no dynamic adaptation across sessions or tasks.

1.3 Motivation

The central motivation for NeuroLearn arises from the persistent mismatch between how educational technology delivers content and how learners naturally process it. Engineering education, in particular, demands the processing of abstract mathematical concepts, algorithmic procedures, and hardware-level design — all of which place high cognitive demands on students. Research consistently shows that when content is presented in a format misaligned with a learner’s preferred modality, cognitive load increases and retention suffers.

Existing adaptive systems have largely solved the problem of *what* to teach (difficulty adaptation) but have left largely unaddressed the equally important question of *how* to teach it (modality adaptation). Furthermore, accessibility has been chronically under-prioritized as a design concern, creating educational inequities for students with visual, auditory, or motor impairments. NeuroLearn is motivated by the conviction that genuine personalization must encompass both presentation modality and accessibility support, unified within a single adaptive architecture rather than implemented as afterthoughts.

1.4 Problem Statement

Current AI-powered educational platforms fail to dynamically adapt the modality of content delivery to individual learner preferences, relying instead on static self-reported

assessments or difficulty-based adaptations that ignore how learners actually interact with content. Furthermore, accessibility support is treated as a supplementary feature rather than an architectural requirement, leaving learners with disabilities at a systematic disadvantage. This project addresses the need for a unified, behavior-driven, accessibility-first adaptive learning framework that continuously infers modality preferences from real-time interaction data and delivers personalized multi-modal content grounded in verified curricular knowledge.

1.5 Objectives

The objectives of the project are:

1. To design and implement a Dynamic Modality Preference Modeling (DMPM) framework that fuses psychometric VARK assessments with real-time behavioral interaction data using exponential decay weighting, achieving superior learner profiling accuracy over static approaches.
2. To develop a modality-aware Retrieval-Augmented Generation (RAG) pipeline that generates curriculum-aligned content across all four VARK modalities — visual animations, audio explanations, structured text summaries, and interactive simulations — grounded in a FAISS vector database of course materials.
3. To integrate accessibility as a core architectural requirement by implementing an accessibility-first adaptation algorithm that automatically routes learners with visual, auditory, or motor impairments to appropriate modalities, ensuring WCAG 2.1 Level AAA compliance throughout the platform.

1.6 Brief Methodology of the Project

The methodology of NeuroLearn follows a three-tier adaptive pipeline. In the first tier (**Learner Profiling**), new users complete a 16-item VARK questionnaire that generates an initial modality preference vector $\mathbf{p}_{\text{VARK}} = [v, a, r, k]$. As the user interacts with the platform, a behavioral profile vector is computed by weighting interaction frequency and duration across content modalities. These two signals are fused using the DMPM algorithm:

$$\mathbf{p}_{\text{final}} = \alpha(n) \mathbf{p}_{\text{VARK}} + (1 - \alpha(n)) \mathbf{p}_{\text{behavior}}, \quad \alpha(n) = e^{-\lambda n}$$

where n is the session count and $\lambda = 0.15$ was empirically determined through pilot testing.

In the second tier (**Content Generation**), the dominant modality is identified and a user query is processed through the RAG pipeline: BERT encodes the query, FAISS retrieves the top- k most relevant passages from the course knowledge base (cosine similarity ≥ 0.6), and Gemini generates modality-specific content — Manim animation scripts for visual learners, BART-summarized audio via Google Cloud TTS for auditory learners, detailed text explanations for reading/writing learners, and interactive sandbox simulations for kinesthetic learners.

In the third tier (**Accessibility Override**), a supervisory algorithm checks for declared accessibility flags before routing content. Visually impaired users are always routed to audio-centric modalities with full screen reader support, regardless of inferred preferences. Hearing-impaired users receive visual-dominant content with complete captions and transcripts. Motor-impaired users are served via full keyboard navigation and voice control. Profile refinement occurs continuously as new interaction data is collected, progressively improving personalization accuracy over sessions.

1.7 Assumptions Made / Constraints of the Project

The following assumptions and constraints apply to this project:

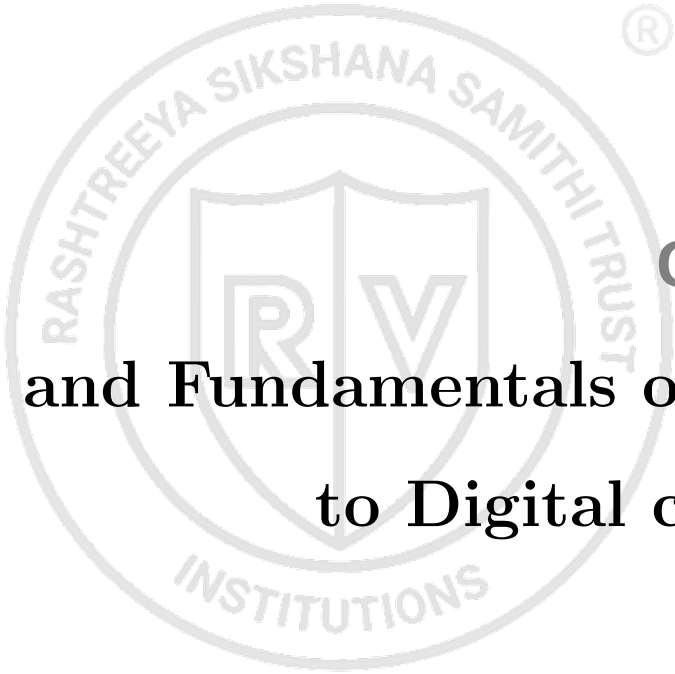
- The system assumes learners have access to a reliable internet connection and a modern web browser, as content delivery relies on cloud-based TTS, animation rendering, and API calls.
- The VARK questionnaire provides a reasonable initialization prior; the system is designed to recover from inaccurate self-assessments through behavioral correction over subsequent sessions.
- The FAISS vector knowledge base is pre-populated with RVCE course lecture notes for the four subjects used in evaluation; the system's content quality is bounded by the quality of these source materials.
- The exponential decay parameter $\lambda = 0.15$ was empirically optimized on a pilot cohort of 20 students; its generalizability to significantly different educational contexts may require re-tuning.

- Visual animation generation (Manim) has higher latency (mean 5.4s) than other modalities; progressive loading and caching strategies are implemented but a stable network is assumed.
- The study is validated on undergraduate engineering students; generalizability to other disciplines (humanities, K-12, professional education) has not been empirically tested.

1.8 Organization of the Report

This report is organized as follows:

- Chapter 1 introduces the project, presents the literature review, defines the problem statement and objectives, outlines the methodology, and lists assumptions and constraints.
- Chapter 2 discusses the system design and architecture, detailing the three-tier adaptive pipeline, the hybrid learner profiling mechanism, and the multi-modal content generation engine.
- Chapter 3 discusses the algorithmic framework, presenting the three core algorithms: Dynamic Modality Preference Modeling (DMPM), Modality-Aware RAG Generation, and Accessibility-First Adaptation.
- Chapter 4 discusses the implementation details, covering the full technical stack, privacy and security mechanisms, and deployment infrastructure.
- Chapter 5 discusses the experimental results, including profiling accuracy, learning outcomes, engagement metrics, ablation studies, and accessibility impact assessments.
- Chapter 6 concludes the report with a summary of contributions, discussion of limitations, and directions for future research.



Chapter 2

Theory and Fundamentals of Analog to Digital converter

CHAPTER 2

THEORY AND FUNDAMENTALS OF ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters 2 to end with "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

2.1 Contents of this chapter

This chapter should discuss about the prerequisite learnings before the execution of the project. Organise and elaborate the theory and necessary fundamentals required for the execution of the project. You can use \subsections and subsubsections in this chapter.

2.2 Contents of this chapter

If a specific programming language is required for the project, a section can be allotted in this chapter to discuss it.

2.3 Contents of this chapter

Tools used could be another possible section to discuss about the software tools used in the work.

2.4 Contents of this chapter

The details in this chapter can be added in consultation with the project guide. For an internship based projects, subsections can be modified accordingly.

2.5 Use of Acronyms and Glossaries

Acronyms are nothing but the short form of regular repeated word. Say for example, you have a repeat word "Integrated Circuits" and you want to use a short form for it as "IC". For which you have to first define the word and use it wherever you wanted to refer it.

First, let's look at the definition, which has to be entered in `Glossaries.tex` under `CoverPages` directory.

```
%\newacronym{<Ref>}{<Short-Form>}{<Expanded word>}  
\newacronym{ic}{IC}{Integrated Circuits}
```

In order to use the defined acronym, use the commands `\gls{<Ref>}` as shown below

As an example, call the definition with `\gls{ic}` and the outcome of it is reflected as, Integrated Circuit (IC).

Note: For the First time, the expanded form appears along with the Short-form definition inside parenthesis. But when the `\gls{}` is repeated, only Short-form appears inside the parenthesis.

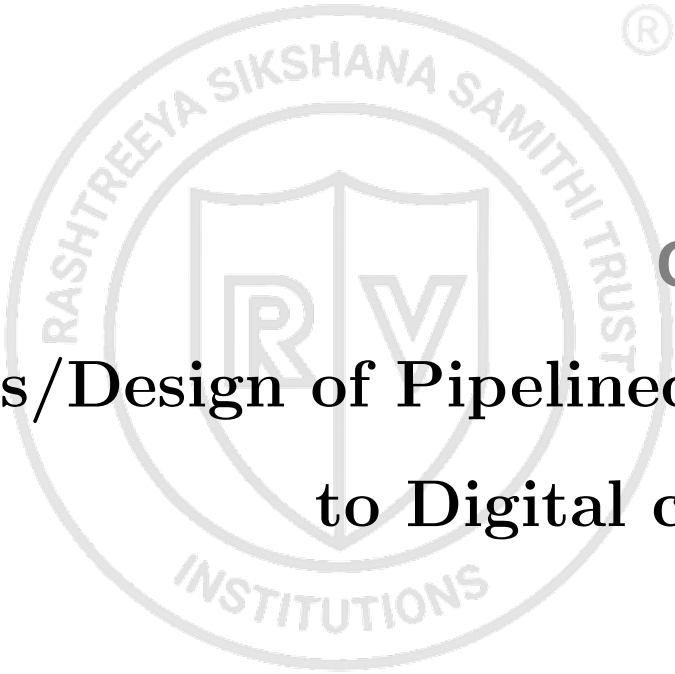
Now, let's look at the definition of symbols. Follow the syntax to define the symbol first, inside `Glossaries.tex` under `CoverPages` directory.

```
%\newglossaryentry{<Ref>}{name=<Symbol>, description={<description about the symbol>}}  
\newglossaryentry{rc}{name=$\tau$, description={Time constant}, type=symbolList}
```

As an example, the rate of change is defined with `\gls{rc}` and the outcome of it is reflected as, the rate of change is defined with τ .

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 3

Analysis/Design of Pipelined Analog to Digital converter

CHAPTER 3

ANALYSIS/DESIGN OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

3.1 Contents of this Chapter

This chapter should contain the following sections and subsections in detail.

1. Specifications for the Design
2. Pre analysis work for the design or Models used
3. Design methodology in detail
4. Design Equations
5. Experimental techniques (if any)

Apart from the aforementioned sections, you can add sections as per the requirements of the project in consultation with your guide.

3.2 Paraphrasing

When you paraphrase a written passage, you rewrite it to state the essential ideas in your own words. Because you do not quote your source word for word when paraphrasing, it is unnecessary to enclose the paraphrased material in quotation marks. However, the paraphrased material must be properly referenced because the ideas are taken from someone else whether or not the words are identical.

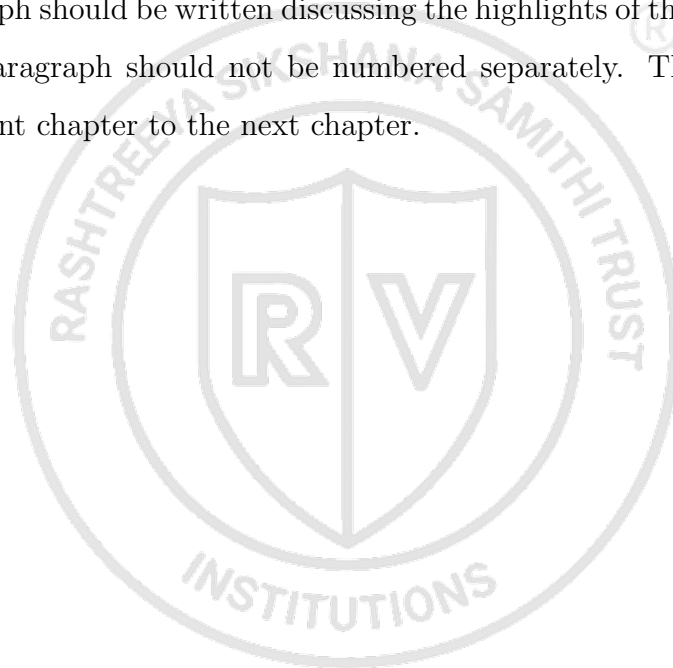
Ordinarily, the majority of the notes you take during the research phase of writing your report will paraphrase the original material. Paraphrase only the essential ideas. Strive to put original ideas into your own words without distorting them."

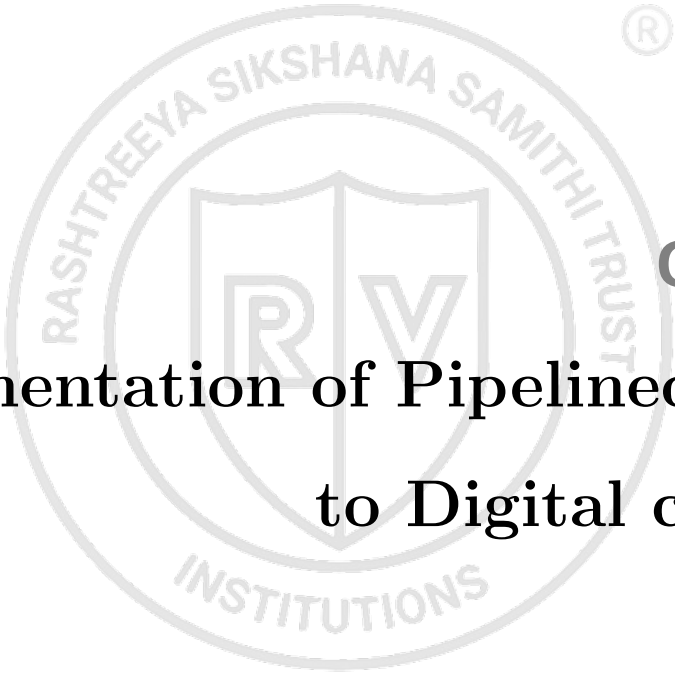
3.3 Quotations

When you have borrowed words, facts, or idea of any kind from someone else's work, acknowledge your debt by giving your source credit in footnote (or in running text as cited reference). Otherwise, you will be guilty of plagiarism. Also, be sure you have represented the original material honestly and accurately. Direct word to word quotations are enclosed in quotation marks.”

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 4

Implementation of Pipelined Analog to Digital converter

CHAPTER 4

IMPLEMENTATION OF PIPELINED ANALOG TO DIGITAL CONVERTER

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with....". Instead you should bring in the highlights of the chapter in the introduction paragraph.

4.1 Contents of this chapter

This chapter should elaborate the following in detail.

1. Implementation details for hardware based projects
2. Top level Design for software based projects

You can add sections and sub sections to elaborate your project work done.

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.



Chapter 5

Results & Discussions

CHAPTER 5

RESULTS & DISCUSSIONS

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

5.1 Contents of this chapter

All the results obtained for your objectives should be discussed in this chapter. This chapter should contain the following sections as per the project.

1. Simulation results
2. Experimental results
3. Performance Comparison
4. Inferences drawn from the results obtained

All the figures should be properly explained by bringing the scenarios of the design done in the project. A detailed discussion of results obtained should be done in this chapter.

5.2 Tables in thesis

- All Table Caption should be in Sentence Case, TNR 10 Pt. It should be of the Format:
 - Table 1.1 Results of the experiment ... (Centered)
- It should be cited as Table 1.1.
- Caption should appear above the Table.
- Table Header and the entries should be of Font TNR 10 Pt, Justified.
- For wider Table, the page orientation can be Landscape.

- For Larger Table, it can run to pages and the header should be repeated for each page of the Table.
- Table must be adjusted to fit in the page and no single row is left out for a new page.

Sample Table 5.1 is given below for your reference,

Table 5.1: Country List

Country Name or Area Name	ISO ALPHA 2 Code	ISO ALPHA 3 Code	ISO numeric Code
Afghanistan	AF	AFG	004
Aland Islands	AX	ALA	248
Albania	AL	ALB	008
Algeria	DZ	DZA	012
American Samoa	AS	ASM	016
Andorra	AD	AND	020
Angola	AO	AGO	024

5.3 Math equation in thesis

All equation should be written using equation editor or using an equivalent tool.

- Equations should be numbered as : 1.1, 1.2 ...
- Equation should be Centered, 12 Pt, TNR.
- Equation number should be right Justified
- It should be cited as Eqn. 1.1.
- If the sentence starts by citing an equation, then it should be written as Equation 1.1 For example, Equation 5.1 states the Pythagoras theorem.

For example in Eqn. 5.1, The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning the next equation has no integer solutions:

$$x^n + y^n = z^n \quad (5.1)$$

The mass-energy equivalence is described by the famous equation in Eqn. 5.2

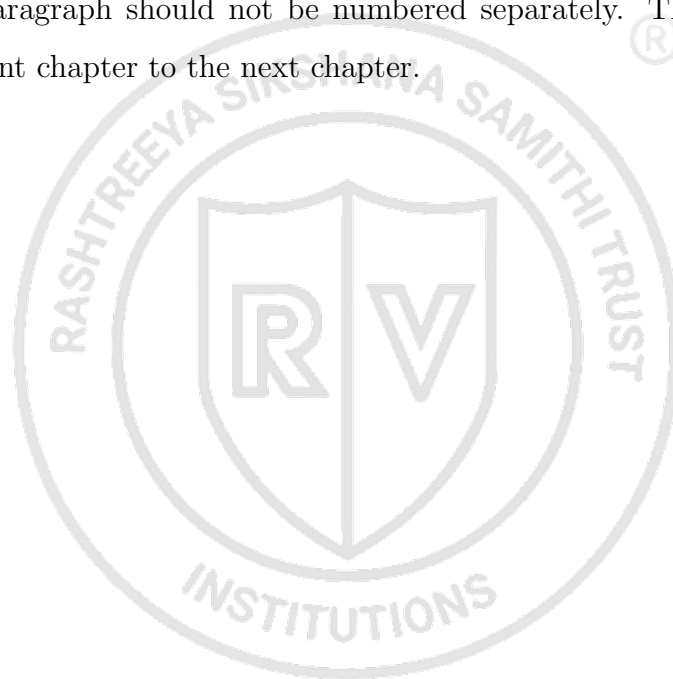
$$E = mc^2 \quad (5.2)$$

discovered in 1905 by Albert Einstein.

5.4 Interpretation of results

The chapters should not end with figures, instead bring the paragraph explaining about the figure at the end followed by a summary paragraph.

After elaborating the various sections of the chapter (From Chapter 2 onwards), a summary paragraph should be written discussing the highlights of that particular chapter. This summary paragraph should not be numbered separately. This paragraph should connect the present chapter to the next chapter.





Chapter 6

Conclusion and Future Scope

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

Every chapter should start with an introduction paragraph. This paragraph should brief about the flow of the chapter. This introduction can be limited within 4 to 5 sentences. The chapter heading should be appropriately modified (a sample heading is shown for this chapter). But don't start the introduction paragraph in the chapters like "This chapter deals with...". Instead you should bring in the highlights of the chapter in the introduction paragraph.

6.1 Conclusion

This chapter should not contain an introduction paragraph like other chapters. You can directly write conclusion of the work done under this section. Typically this section can have 3 to 4 paragraphs.

First paragraph should bring in the scenario of the project and every objective should be explained here.

Second paragraph should say how the objectives are implemented and achieved.

Last paragraph should draw the conclusions from each objective with quantitative results, performance improvement etc.

6.2 Future Scope

Briefly discuss the constraints and limitations of the project and state the possibilities of extending the work in future.

6.3 Learning Outcomes of the Project

- List the learning outcomes here
- List a minimum of 5 learning outcomes



Appendix A

Code

APPENDIX A

CODE

A.1 First Appendix

You can use `tcbllisting` for creating the code snippets. The following example illustrates how one can customize the `tcbllisting` to achieve the `tcl` script. Similarly, one can use it for other programming language listing, including HDL.

```
# Since our design has a clock with name clk,
## specify that name under [get_port ]
create_clock -period 40 -waveform {0 20} [get_ports clk]

# Setting a 'delay' on the clock:
set_clock_latency 0.3 clk

# Setting up constraints on your I/P and O/P pins
set_input_delay 2.0 -clock clk [all_inputs]
set_output_delay 1.65 -clock clk [all_outputs]

# Set realistic 'loads' on each output pin
set_load 0.1 [all_outputs]

# Set 'maximum' fanin and fan-out for the input and output pins
set_max_fanout 1 [all_inputs]
set_fanout_load 8 [all_outputs]
```